Bài 4: Photosynthesis in Plants

English Vocabulary Worksheet · Long-Lesson Mode

Long-lesson mode: all exercise counts doubled.

THEORY NOTES

I. Overview of Photosynthesis

1. Concept

Photosynthesis is the process by which plant chloroplasts absorb and use light energy to convert CO_2 and water into the organic compound glucose ($C_6H_{12}O_6$) while simultaneously releasing O_2.

Overall equation:

$$6CO_2 + 12H_2O \xrightarrow{\text{(light, chloroplast)}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

2. Role of Photosynthesis

- Provides diverse organic nutrients for virtually all organisms on Earth.
- Supplies raw materials and energy fuels for industry; fossil fuels and other photosynthesis products account for approximately 90 % of human energy consumption.
- Provides large amounts of chemical energy to sustain biological activity across the biosphere.
- Maintains the atmospheric O_2/CO_2 balance through continuous O_2 release.

Keywords: *organic matter · fossil fuel · oxygen · CO_2 balance · biosphere energy*

3. Photosynthetic Pigments

Two main groups: chlorophylls and carotenoids.

Chlorophylls: ~10 types known; chlorophyll a is the primary pigment. It is the only pigment that converts light energy into chemical energy at the reaction centre — without it, photosynthesis cannot occur. Chlorophyll a absorbs primarily red and blue-violet light.

Carotenoids: create yellow, red, and orange colours in leaves, flowers, and fruits. Include xanthophylls (oxygen-containing) and carotenes (oxygen-free). β-carotene is a precursor of vitamin A.

All pigments are distributed on thylakoid membranes and form a light-harvesting network that funnels absorbed energy through carotenoids and chlorophyll b to chlorophyll a at the reaction centre, which converts it into chemical energy (ATP and NADPH).

Keywords: *chlorophyll a · carotenoid · thylakoid membrane · reaction centre · ATP · NADPH*

II. Process of Photosynthesis in Plants

Photosynthesis occurs in the chloroplast in two stages: the light phase (light-dependent reactions) and the dark phase (light-independent reactions / Calvin cycle).

1. Light Phase

Location: thylakoid membranes. Occurs identically in all plant groups (C_3, C_4, CAM).

Process: Chlorophyll absorbs photons → electrons excited and emitted. Water photolysis replaces them:

$$2H_2O \rightarrow 4H^+ + 4e^- + O_2$$

H^+ and electrons synthesise ATP and reduce $NADP^+$ to NADPH. O_2 is released.

Products: O_2, ATP, NADPH

Keywords: *water photolysis · thylakoid · ATP · NADPH · O_2 · excited electron*

2. Dark Phase (Light-Independent Reactions)

Location: stroma of the chloroplast. Uses ATP and NADPH from the light phase plus enzymes to fix CO_2.

a) C_3 Pathway (Calvin Cycle): Described by Calvin and Benson (1951). Dominant in temperate and sub-tropical plants: rice, potato, cassava, most vegetables and legumes.

RuBP (5C) + CO_2 → 3-PGA (3C) → G3P (3C) [using ATP + NADPH] → glucose / other organics; remaining G3P regenerates RuBP.

b) C_4 Pathway: Discovered by Hatch and Slack (1965). Tropical/subtropical plants: sugarcane, maize, sorghum, Bermuda grass.

Mesophyll cells: PEP (3C) + CO_2 → OAA (4C). OAA → bundle sheath cells → releases CO_2 → Calvin cycle.

c) CAM Pathway: Desert/water-limited plants: cactus, pineapple, stonecrop.

Night: stomata open; CO_2 fixed via PEP → OAA → malate (stored). Day: stomata closed; malate releases CO_2 → Calvin cycle.

Keywords: *RuBP · 3-PGA · G3P · PEP · OAA · mesophyll · bundle sheath · CAM · stroma*

3. Adaptation of C_4 and CAM Plants

In hot or dry conditions, C_4 and CAM plants have a preliminary CO_2-fixation step that concentrates CO_2 for the Calvin cycle, enabling high efficiency under stress.

PEP-carboxylase: has very high CO_2 affinity → fixes CO_2 efficiently even when stomata are partly closed (low intercellular CO_2).

C_4 plants accumulate CO_2 in mesophyll cells → reliable CO_2 supply for Calvin cycle → high yields. CAM plants open stomata only at night → minimise water loss.

Keywords: *PEP-carboxylase · CO_2 affinity · water conservation · high yield*

4. Role of Photosynthetic Products

G3P (glyceraldehyde 3-phosphate) is the primary photosynthetic product in all three plant groups and serves as a universal building block.

G3P → Glucose → Carbohydrates

G3P → Amino acids → Proteins

G3P → Fatty acids → Lipids

Keywords: *G3P · glucose · carbohydrate · amino acid · protein · fatty acid · lipid*

III. Influence of External Factors on Photosynthesis

1. Light

Intensity: Light compensation point (I_o): intensity where photosynthesis rate = respiration rate. Shade plants have a lower compensation point than sun plants.

Light saturation point (I_m): intensity at which photosynthesis peaks; further increases yield no additional gain in rate.

Composition: Visible spectrum (400–700 nm): violet, blue, green, yellow, orange, red. Photosynthesis is most effective in red and blue-violet light; red light drives the highest rate.

Blue-violet light promotes amino acid and protein synthesis; red light drives carbohydrate formation. Light composition changes throughout the day.

Keywords: *compensation point · saturation point · red light · blue-violet light · sun plant · shade plant*

2. CO_2

Within limits, higher CO_2 → higher photosynthesis rate. CO_2 above ~0.2 % is toxic to plants. CO_2 compensation point: lower in C_4 and CAM than in C_3. CO_2 saturation point for crop plants: approximately 0.06–0.1 %.

Keywords: *CO_2 compensation point · CO_2 saturation point · CO_2 toxicity*

3. Temperature

Tropical plants begin photosynthesis at 5–7 °C; cold/temperate plants begin at −15 to −5 °C. Rate rises to an optimum then falls. Tropical optimum: 25–30 °C; temperate optimum: 8–15 °C.

Keywords: *optimum temperature · tropical crops · temperate crops*

IV. Photosynthesis and Crop Yield

1. Relationship Between Photosynthesis and Crop Yield

Photosynthesis produces 90–95 % of crop dry matter; mineral nutrition accounts for the remaining 5–10 %. Photosynthesis is therefore the primary determinant of agricultural yield.

Keywords: *dry matter · crop yield · mineral nutrition*

2. Technical Measures to Improve Crop Yield

Fertilisation: Potassium for tuber crops increases starch and sugar translocation to storage organs. Nitrogen expands leaf area, boosting photosynthesis efficiency. Legumes benefit from nitrogen for protein synthesis.

Water: A raw material of photosynthesis; adequate supply—especially at the reproductive stage—increases efficiency and prevents grain/fruit failure.

Planting time: Match planting time to temperature and light requirements. Example: potato tuber formation is optimal at 18–23 °C.

Planting density: Too dense → insufficient light per plant → lower photosynthesis rate and yield.

Keywords: *potassium · nitrogen · leaf area · water · planting time · planting density · yield*

PART 1 · Match the Terms (A–T)

Write the correct letter from the Definitions list in the Answer column. Each letter is used exactly once.

#	Term	Answer
1	Photosynthesis	
2	Chlorophyll a	
3	Carotenoid	
4	Thylakoid membrane	
5	Light phase (light-dependent reactions)	
6	Water photolysis	
7	ATP	
8	NADPH	
9	Dark phase / Calvin cycle	
10	RuBP	
11	3-PGA	
12	G3P	
13	C_4 pathway	
14	PEP	
15	CAM pathway	
16	Light compensation point	
17	Light saturation point	
18	CO_2 compensation point	
19	Dry matter	
20	PEP-carboxylase	

Definitions

Letter	Definition
A	The three-carbon carbohydrate produced in the Calvin cycle that serves as the building block for glucose, amino acids, and other organic molecules
B	A CO_2-fixation strategy used by desert and water-limited plants that separates nighttime carbon fixation from daytime Calvin cycle activity
C	The light intensity above which further increases no longer raise the rate of photosynthesis (maximum photosynthesis is reached)
D	The light-driven splitting of water molecules that releases protons, electrons, and molecular oxygen as a by-product
E	The minimum CO_2 concentration at which the rate of photosynthesis exactly equals the rate of cellular respiration
F	The internal membrane system of the chloroplast on which photosynthetic pigments are embedded and where the light-phase reactions occur
G	The three-carbon molecule that acts as the primary CO_2 acceptor in C_4 and CAM plants; catalysed by PEP-carboxylase
H	A CO_2-fixation route in which PEP is the first CO_2 acceptor in mesophyll cells and oxaloacetic acid (OAA) is the first four-carbon product
I	An energy-carrying molecule synthesised during the light phase that provides chemical energy to drive the reactions of the dark phase
J	The five-carbon molecule that accepts CO_2 at the start of each turn of the Calvin cycle, forming the first stable three-carbon product
K	The stage of photosynthesis occurring in the stroma that uses ATP and NADPH to fix CO_2 into organic molecules; also called the Calvin or C_3 cycle
L	The first stable three-carbon product formed when RuBP reacts with CO_2 in the Calvin cycle; subsequently reduced to G3P

Letter	Definition
M	A group of accessory pigments that create yellow, red, and orange colours in leaves, flowers, fruits, and roots; includes xanthophylls and carotenes
N	The process by which plant chloroplasts use light energy to convert CO_2 and water into glucose while simultaneously releasing molecular oxygen
O	The stage of photosynthesis on thylakoid membranes requiring light, which produces ATP, NADPH, and O_2 through water photolysis and electron transport
P	A reducing agent produced during the light phase that donates electrons and hydrogen to reduce 3-PGA into G3P in the Calvin cycle
Q	The primary photosynthetic pigment that absorbs red and blue-violet light and is the only pigment that converts light energy to chemical energy at the reaction centre
R	The enzyme with exceptionally high CO_2 affinity that catalyses the first carbon-fixation step in C_4 and CAM plants, enabling fixation at very low CO_2 concentrations
S	The fraction of crop biomass derived from photosynthesis, constituting approximately 90–95 % of total plant dry mass in agricultural crops
T	The light intensity at which the rate of photosynthesis equals the rate of cellular respiration; shade plants have a lower value than sun plants

PART 2 · Drag & Drop

Fill each numbered blank with a word or phrase from the Word Bank. Each item in the Word Bank is used exactly once per passage.

Passage A — Overview and Role of Photosynthesis

Photosynthesis is the process by which plant (1)___ absorb and use light energy to convert (2)___ and (3)___ into (4)___, releasing (5)___ as a by-product. The (6)___ matter produced sustains the nutritional needs of nearly all living organisms.

Word Bank: *chloroplasts* · *carbon dioxide* · *water* · *glucose* · *oxygen* · *organic*

Passage B — Photosynthetic Pigments

Photosynthetic pigments are located on the (1)___ membrane inside the chloroplast. When a pigment molecule absorbs a photon, its electrons become (2)___. Accessory pigments such as (3)___ collect light and transfer its (4)___ toward chlorophyll b, which relays it to (5)___ at the (6)___, where light energy is converted into chemical form.

Word Bank: *thylakoid* · *excited* · *carotenoids* · *energy* · *chlorophyll a* · *reaction centre*

Passage C — The Light Phase

During the light phase, (1)___ of water molecules occurs: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The (2)___ ($H^+$) released contribute to synthesising (3)___. The (4)___ generated replace those lost from chlorophyll a and reduce $NADP^+$ to (5)___. The (6)___ liberated during this process exits through stomata into the atmosphere.

Word Bank: *photolysis* · *protons* · *ATP* · *electrons* · *NADPH* · *oxygen*

Passage D — The Calvin Cycle (C_3 Pathway)

The (1)___ takes place in the (2)___ of the chloroplast. The five-carbon molecule (3)___ accepts CO_2 to produce the three-carbon compound (4)___. ATP and NADPH then reduce 3-PGA to (5)___, which the plant uses to synthesise (6)___ and other organic molecules, or to regenerate RuBP for the next cycle turn.

Word Bank: *Calvin cycle* · *stroma* · *RuBP* · *3-PGA* · *G3P* · *glucose*

Passage E — C_4 and CAM Pathways

In C_4 plants, CO_2 is first fixed in (1)___ cells, where PEP-carboxylase attaches it to (2)___ to yield (3)___, a four-carbon compound. This compound migrates to (4)___ cells, where CO_2 is released into the Calvin cycle. In (5)___ plants, (6)___ remain closed during the day and open at night, allowing CO_2 uptake while minimising water loss in arid environments.

Word Bank: *mesophyll* · *PEP* · *oxaloacetic acid* · *bundle sheath* · *CAM* · *stomata*

Passage F — External Factors and Crop Yield

Photosynthesis produces 90–95 % of the (1)___ in crop plants, making it the primary

yield determinant. Among external factors, (2)___ sets the optimal reaction rate; most tropical crops perform best at 25–30 °C. Adequate (3)___ application—especially nitrogen—enlarges leaf area and raises photosynthetic efficiency. The CO_2 (4)___ point marks where photosynthesis rate just equals respiration rate. The CO_2 (5)___ point is where the maximum photosynthesis rate is achieved. Choosing the right planting time ensures optimal (6)___ intensity for each crop variety.

Word Bank: *dry matter* · *temperature* · *fertiliser* · *compensation* · *saturation* · *light*

PART 3 · Reading + Multiple-Choice Questions

Read each passage and choose the best answer (A, B, C, or D).

Passage 1 (Q1–Q5) — Overview and Role of Photosynthesis

> Photosynthesis is the process by which plant chloroplasts capture light energy and use it to transform carbon dioxide and water into glucose ($C_6H_{12}O_6$), simultaneously releasing molecular oxygen. The overall equation is: $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$. The process occurs in two main stages—the light-dependent phase and the light-independent phase—and takes place entirely within the chloroplast. Photosynthesis provides organic matter for virtually all organisms, supplies raw materials and energy for human industries (fossil fuels derived from photosynthetic organisms account for roughly 90 % of human energy consumption), and maintains the atmospheric oxygen–carbon dioxide balance.

1. In the photosynthesis equation, what is the coefficient of water on the reactant side?

A) 6
B) 12
C) 18
D) 24

2. According to the passage, what fraction of human energy consumption derives from photosynthesis products?

A) About 50 %
B) About 70 %
C) About 80 %
D) About 90 %

3. The phrase "light-independent phase" most nearly means a stage that

A) does not require direct light energy to proceed
B) releases visible light as a reaction by-product
C) stops permanently whenever light is removed
D) takes place on the outer chloroplast surface

4. Which best describes the role of photosynthesis in maintaining the atmosphere?

A) It removes nitrogen from the air and fixes it in soil
B) It maintains the balance between O_2 and CO_2 levels
C) It converts all atmospheric CO_2 into water vapour
D) It raises the concentration of greenhouse gases steadily

5. Based on the passage, which statement about the location of photosynthesis is accurate?

A) Photosynthesis occurs entirely inside the mitochondrion
B) Both stages of photosynthesis take place within the chloroplast
C) Only the light phase takes place inside the chloroplast
D) The light-independent phase occurs directly in the cytoplasm

Passage 2 (Q6–Q10) — Photosynthetic Pigments

> Plants contain two main groups of photosynthetic pigments: chlorophylls and carotenoids. Chlorophylls give leaves and green parts their colour. Among approximately ten types, chlorophyll a is most important: it is the only pigment capable of converting light energy into chemical energy at the reaction centre; without it, photosynthesis cannot occur. Chlorophyll a absorbs primarily red and blue-violet light. Carotenoids—including xanthophylls (oxygen-containing) and carotenes (oxygen-free)—create yellow, red, and orange colours. Beta-carotene serves as a precursor molecule for vitamin A synthesis.

All pigments reside on the thylakoid membrane, forming a light-harvesting network that channels energy to chlorophyll a at the reaction centre where it is converted to ATP and NADPH.

6. Which pigment is described as indispensable for photosynthesis to occur?

A) Chlorophyll b
B) Chlorophyll a
C) Xanthophyll
D) Beta-carotene

7. The word "indispensable" in context is closest in meaning to

A) replaceable
B) available
C) essential
D) visible

8. According to the passage, beta-carotene primarily functions as

A) an oxygen-carrying molecule in the light phase
B) a precursor molecule for vitamin A synthesis
C) the primary light absorber at the reaction centre
D) a molecule that absorbs red and blue-violet light

9. What is the primary role of xanthophylls and carotenes in the light-harvesting network?

A) Convert light energy directly to chemical energy here
B) Absorb light energy and transfer it toward chlorophyll a
C) Split water molecules to release molecular oxygen inside
D) Synthesise ATP and NADPH for use in the dark phase

10. The passage implies that the location of pigments on the thylakoid membrane is significant because

A) it allows pigments to be exposed directly to CO_2
B) it prevents chlorophyll a from absorbing red light
C) it enables efficient energy transfer between pigment molecules
D) it keeps chlorophyll b isolated from all carotenoid molecules

Passage 3 (Q11–Q15) — The Light Phase

The light phase of photosynthesis occurs on the thylakoid membranes. When chlorophyll molecules absorb photons, their electrons are excited and leave the molecule. These electrons are replaced by electrons from the splitting of water—a process called photolysis: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The protons ($H^+$) and electrons produced participate in synthesising ATP and reducing $NADP^+$ to NADPH. Oxygen is released into the atmosphere as a by-product. The three products of the light phase—ATP, NADPH, and O_2—are essential: ATP and NADPH power the subsequent dark phase, while O_2 exits through stomata into the surrounding air.

11. Where does the light phase of photosynthesis take place?

A) In the stroma of the chloroplast matrix
B) On the outer envelope of the chloroplast
C) On the thylakoid membranes inside the chloroplast
D) In the cytoplasm surrounding the chloroplast

12. In the equation $2H_2O \rightarrow 4H^+ + 4e^- + O_2$, what role does oxygen play?

A) A reactant consumed during carbon fixation reactions
B) A by-product released from the water-splitting step
C) An electron donor that directly replaces excited electrons
D) A molecule that drives ATP synthesis in the dark phase

13. According to the passage, which event occurs first during the light phase?

A) NADPH reduces 3-PGA to G3P inside the stroma
B) Water photolysis provides electrons to replenish chlorophyll
C) CO_2 is accepted by RuBP to form a three-carbon product
D) G3P exits the Calvin cycle to be converted to glucose

14. The term "photolysis" as used in the passage most nearly means

A) the synthesis of new photosynthetic pigment molecules
B) the breakdown of water molecules using absorbed light
C) the absorption of photons by carotenoid accessory pigments
D) the transfer of electrons from chlorophyll directly to RuBP

15. Which of the following is NOT a direct product of the light phase?

A) ATP
B) NADPH
C) O_2
D) G3P

Passage 4 (Q16–Q20) — The Calvin Cycle (C_3 Pathway)

> The Calvin cycle, also called the C_3 pathway, was described by Calvin and Benson in 1951. It takes place in the stroma of the chloroplast and is the dominant CO_2-fixation route in temperate and sub-tropical plants—such as rice, potato, cassava, most vegetables and legumes. In this cycle, CO_2 is accepted by ribulose 1,5-bisphosphate (RuBP), a five-carbon molecule, producing two molecules of 3-phosphoglyceric acid (3-PGA), a three-carbon compound—which is why the pathway is named C_3. ATP and NADPH from the light phase reduce 3-PGA to glyceraldehyde 3-phosphate (G3P). A portion of G3P leaves the cycle to build glucose and other organic compounds; the rest regenerates RuBP to continue the cycle.

16. The Calvin cycle is called the C_3 pathway because

A) it uses three specific enzymes to complete all steps
B) the first stable product of CO_2 fixation is three-carbon
C) three CO_2 molecules are used in each complete cycle turn
D) it was discovered by exactly three scientists in 1951

17. Which molecule is the primary CO_2 acceptor in the Calvin cycle?

A) G3P
B) 3-PGA
C) RuBP
D) ATP

18. Which plants primarily rely on the C_3 pathway, according to the passage?

A) Sugarcane and maize in tropical regions
B) Rice and potato in temperate and sub-tropical regions
C) Cactus and pineapple in arid desert environments
D) Bermuda grass and sorghum in hot subtropical savannahs

19. What are the two fates of G3P molecules produced in the Calvin cycle?

A) All of it is used immediately to regenerate RuBP
B) It is converted back to CO_2 and water in the stroma
C) Part builds organic compounds; the rest regenerates RuBP
D) It is moved to the thylakoid membrane to produce ATP

20. The phrase "dominant CO_2-fixation route" most nearly means

A) the only known pathway for carbon fixation in plants
B) the most widespread carbon-fixation pathway across plant species

C) the fastest method for incorporating CO_2 into plant cells
D) the pathway yielding the most ATP per molecule of CO_2

Passage 5 (Q21–Q25) — C_4 and CAM Pathways

> In 1965, Hatch and Slack discovered that some plants—including sugarcane, maize, sorghum, and Bermuda grass—fix CO_2 through the C_4 pathway. In mesophyll cells, phosphoenol pyruvate (PEP) acts as the first CO_2 acceptor, and the initial four-carbon product is oxaloacetic acid (OAA). OAA migrates to bundle sheath cells, releases CO_2, which enters the Calvin cycle. PEP-carboxylase has very high CO_2 affinity, enabling fixation even at low concentrations. CAM plants (cacti, pineapples) use a different strategy: stomata close during the day and open at night. At night CO_2 is fixed via PEP to form malate; during the day, stored malate releases CO_2 into the Calvin cycle while stomata remain closed.

21. What is the first CO_2 acceptor molecule in C_4 plants?
A) RuBP
B) 3-PGA
C) PEP
D) OAA

22. The C_4 pathway is named because
A) it uses four different enzymes to fix CO_2
B) the initial CO_2-fixation product is a four-carbon molecule
C) the process occurs in four structurally separate compartments
D) it requires exactly four ATP molecules per CO_2 fixed

23. Which best explains why C_4 plants are better adapted to hot, dry conditions than C_3?
A) They carry out photosynthesis without needing any chlorophyll
B) PEP-carboxylase fixes CO_2 efficiently even when stomata are partly closed
C) C_4 plants do not require water molecules for the light-phase reactions
D) C_4 plants release significantly more oxygen per unit leaf area

24. In CAM plants, when do stomata open to allow CO_2 to enter the leaf?
A) During midday when sunlight intensity is the highest
B) At dawn when the ambient temperature is at its lowest
C) At night when the risk of water loss is minimised
D) Continuously throughout the entire day and night cycle

25. According to the passage, OAA moves to bundle sheath cells in C_4 plants in order to
A) absorb additional light energy to power the light phase
B) release CO_2 that then enters the Calvin cycle directly
C) store carbon compounds at night until the plant needs energy
D) convert back to PEP molecules for reuse in mesophyll cells

Passage 6 (Q26–Q30) — External Factors and Crop Yield

> Several external factors govern the rate of photosynthesis. Light is fundamental: below the light compensation point, respiration exceeds photosynthesis and the plant cannot accumulate organic matter; above the light saturation point, more light yields no further gain. CO_2 is another key factor: as CO_2 rises, photosynthesis increases until the CO_2 saturation point, above which further increases have no effect (very high concentrations are toxic). Temperature also matters: most tropical crops reach their maximum rate at 25–30 °C, whereas temperate-zone crops peak at 8–15 °C. Because photosynthesis produces 90–95 % of crop dry matter, agronomists optimise light, CO_2, temperature, water, and fertiliser supply through targeted planting schedules and inputs to maximise yield.

26. Below the light compensation point, which of the following occurs?

A) Photosynthesis and respiration rates are exactly equal
B) The plant accumulates organic matter faster than it uses it
C) Respiration rate exceeds the rate of photosynthesis
D) Photosynthesis stops permanently without additional fertiliser

27. What happens if CO_2 concentration rises above the saturation point?

A) Photosynthesis rate continues increasing proportionally with CO_2
B) The plant grows faster due to excess available carbon molecules
C) The photosynthesis rate no longer increases and may decline
D) The light saturation point automatically rises to compensate

28. The phrase "CO_2 saturation point" most nearly means

A) the CO_2 level at which plant cells become saturated with water
B) the CO_2 concentration at which photosynthesis rate is maximum
C) the minimum CO_2 concentration necessary to begin photosynthesis
D) the CO_2 concentration corresponding to the light saturation point

29. Which strategy would most directly benefit a crop farmer in a hot, dry climate?

A) Reducing light intensity by installing shade nets over crops
B) Providing extra CO_2 by ventilating the greenhouses at night
C) Ensuring sufficient water supply so stomata can remain open
D) Reducing the ambient temperature to below 15 °C for crops

30. Based on the passage, what is the most important reason to optimise photosynthesis-related factors?

A) To significantly increase the amount of O_2 in the atmosphere
B) Because photosynthesis accounts for 90–95 % of crop dry matter
C) To reduce the total amount of fertiliser and irrigation needed
D) Because light intensity is the only variable agronomists can control

PART 4 · Final Multiple-Choice Questions (Q1–Q40)

Choose the best answer (A, B, C, or D) for each question.

Section I — Overview and Role of Photosynthesis (Q1–Q6)

1. What are the raw materials consumed in photosynthesis?
- A) Glucose and oxygen
- B) Carbon dioxide and water
- C) ATP and NADPH molecules
- D) Oxygen and glucose molecules

2. In which organelle does photosynthesis take place?
- A) Mitochondrion
- B) Nucleus
- C) Ribosome
- D) Chloroplast

3. Approximately what fraction of human energy consumption comes from photosynthesis products?
- A) About 50 %
- B) About 70 %
- C) About 80 %
- D) About 90 %

4. Which of the following is NOT a direct role of photosynthesis?
- A) Providing organic nutrients for all organisms on Earth
- B) Maintaining the atmospheric O_2 and CO_2 balance
- C) Breaking down organic molecules to release usable energy
- D) Supplying raw materials for various industrial processes

5. Which gas is released by plants into the atmosphere during photosynthesis?
- A) Carbon dioxide
- B) Nitrogen
- C) Hydrogen
- D) Oxygen

6. The overall photosynthesis equation includes 12 molecules of H_2O because
- A) six water molecules are needed to dissolve the six CO_2 reactants
- B) six appear as reactants and six are produced as new products
- C) water is needed in a 2:1 ratio relative to the glucose produced
- D) twelve water molecules together produce twelve oxygen molecules

Section I — Photosynthetic Pigments (Q7–Q12)

7. Which photosynthetic pigment is indispensable for photosynthesis to occur?
- A) Chlorophyll b
- B) Carotene
- C) Chlorophyll a
- D) Xanthophyll

8. Which light wavelengths does chlorophyll a absorb most strongly?
- A) Green and yellow light
- B) Red and blue-violet light
- C) Orange and violet light
- D) Blue and green light

9. Which pigment group includes beta-carotene, a precursor of vitamin A?
- A) Chlorophylls

B) Carotenoids
C) Phycobilins
D) Anthocyanins

10. Where in the chloroplast are photosynthetic pigments located?
A) In the stroma matrix
B) On the thylakoid membrane
C) On the outer chloroplast envelope
D) Inside the nucleus

11. Which statement correctly describes xanthophylls?
A) A type of carotenoid that contains no oxygen atoms
B) A form of chlorophyll that absorbs mainly blue light
C) A carotenoid group that contains oxygen in its structure
D) An enzyme that transfers light energy to the reaction centre

12. The primary function of carotenoids and chlorophyll b in photosynthesis is to
A) split water molecules directly during the light phase
B) synthesise ATP directly within the chloroplast stroma
C) absorb light energy and channel it to chlorophyll a
D) produce molecular oxygen through a reduction reaction

Section II — The Light Phase (Q13–Q18)

13. Which set of molecules are the three direct products of the light phase?
A) CO_2, H_2O, and glucose
B) ATP, NADPH, and O_2
C) G3P, RuBP, and 3-PGA
D) O_2, CO_2, and $NADP^+$

14. During water photolysis, which molecule directly provides the replacement electrons for chlorophyll a?
A) Carbon dioxide
B) ATP
C) NADPH
D) Water

15. What is the main purpose of NADPH produced in the light phase?
A) To absorb extra light energy for the pigment network
B) To reduce 3-PGA into G3P in the dark-phase reactions
C) To split water molecules during the photolysis reaction
D) To transport oxygen out of the chloroplast to stomata

16. The light phase of photosynthesis occurs on which specific structure?
A) The outer membrane of the chloroplast
B) The stroma matrix of the chloroplast
C) The thylakoid membrane
D) The mitochondrial inner membrane

17. In the equation $2H_2O \rightarrow 4H^+ + 4e^- + O_2$, what role do the H^+ ions play?
A) They directly reduce CO_2 to produce glucose molecules
B) They participate in synthesising ATP and producing NADPH
C) They replace electrons lost from RuBP after CO_2 fixation
D) They oxidise carotenoids to generate additional electrons

18. Which statement correctly describes the role of ATP in photosynthesis?
A) It absorbs light to excite electrons in chlorophyll
B) It accepts electrons from water to generate NADPH
C) It provides energy for the light-independent Calvin cycle

D) It acts as the primary CO_2 acceptor in the dark phase

Section II — The Calvin Cycle (C_3 Pathway) (Q19–Q22)

19. The Calvin cycle takes place in which part of the chloroplast?

A) The thylakoid lumen
B) The thylakoid membrane
C) The outer chloroplast envelope
D) The stroma

20. What is the first three-carbon compound formed when CO_2 is fixed onto RuBP?

A) G3P
B) OAA
C) 3-PGA
D) PEP

21. After G3P is produced, what are the two fates of these G3P molecules?

A) All G3P molecules are used immediately to regenerate RuBP
B) Some G3P builds organic compounds; the rest regenerates RuBP
C) All G3P exits the cycle immediately to form glucose only
D) G3P is transported to the thylakoid membrane to make ATP

22. Which statement most accurately distinguishes the Calvin cycle from the light phase?

A) The Calvin cycle requires direct light energy as an input
B) The Calvin cycle uses ATP and NADPH produced by the light phase
C) The Calvin cycle produces O_2 while the light phase consumes it
D) The Calvin cycle occurs on thylakoid membranes inside the stroma

Section II — C_4 and CAM Pathways and Adaptation (Q23–Q28)

23. In C_4 plants, where does the initial CO_2 fixation step take place?

A) Bundle sheath cells
B) Guard cells
C) Mesophyll cells
D) Epidermal cells

24. Why is the C_4 pathway advantageous in hot, arid climates?

A) C_4 plants photosynthesize without needing any chlorophyll
B) PEP-carboxylase fixes CO_2 efficiently even when stomata are partly closed
C) C_4 plants open stomata more widely than C_3 plants in heat
D) C_4 plants have thicker cuticles that effectively reflect excess heat

25. Which of the following plants belongs to the CAM group?

A) Rice
B) Wheat
C) Corn
D) Cactus

26. In CAM plants, when do stomata open to allow CO_2 absorption?

A) Early morning
B) Mid-afternoon
C) At night
D) All day

27. The ability of C_4 plants to concentrate CO_2 in mesophyll cells is important because

A) it allows photosynthesis to occur without needing water
B) it ensures adequate CO_2 for the Calvin cycle under low-CO_2 conditions
C) it enables C_4 plants to produce more ATP in the light phase
D) it prevents carotenoids from degrading at high temperatures

28. Which feature do C_4 and CAM plants share that helps them adapt to harsh conditions?

A) Both keep stomata open all day to maximise CO_2 uptake
B) Both have a preliminary CO_2-fixation cycle before the Calvin cycle
C) Both lack chlorophyll a and rely solely on carotenoids for energy
D) Both fix all carbon exclusively in bundle sheath cells

Section III — Influence of External Factors (Q29–Q34)

29. What happens to photosynthesis rate when light intensity rises above the light saturation point?

A) It increases rapidly with each additional unit of light
B) It decreases sharply and may damage the leaf cells
C) It remains unchanged regardless of further light increase
D) It drops to zero and then begins recovering gradually

30. Which type of plant has a lower light compensation point?

A) Sun plants
B) Shade plants
C) C_4 plants
D) CAM plants

31. At very high CO_2 concentrations (around 0.2 %), what effect does this have on plants?

A) It maximises the photosynthesis rate indefinitely
B) It causes stomata to open wider than usual
C) It becomes toxic and can kill the plant
D) It has no measurable effect on photosynthesis rate

32. The CO_2 compensation point is defined as

A) the CO_2 concentration that maximises photosynthesis rate
B) the CO_2 level at which photosynthesis rate equals respiration rate
C) the minimum light intensity needed to begin photosynthesis
D) the CO_2 concentration at which maximum crop yield is achieved

33. Compared to tropical crops, temperate-zone crops reach peak photosynthesis rate at

A) a higher temperature range
B) a lower temperature range
C) the same temperature range
D) temperatures below 0 °C only

34. Which best explains why red light drives photosynthesis more effectively than blue-violet at equal intensity?

A) Red light has shorter wavelengths that penetrate leaf tissue deeply
B) Blue-violet light destroys chlorophyll a molecules on absorption
C) Photosynthesis rate is highest in the red band where chlorophyll a absorbs most
D) Red light causes stomata to open more widely than blue-violet light does

Section IV — Photosynthesis and Crop Yield (Q35–Q40)

35. What percentage of crop dry matter originates from photosynthesis?

A) 50–60 %
B) 70–80 %
C) 85–90 %
D) 90–95 %

36. The remaining 5–10 % of crop dry matter is primarily attributed to

A) solar energy absorbed by the soil
B) mineral nutrition derived from the soil
C) water absorbed through the root system
D) gas exchange occurring through the stomata

37. Why should crops not be planted at excessively high density?

A) Dense planting increases disease risk but not crop yield
B) Higher density lowers CO_2 below the compensation point
C) Excessive density reduces light per plant, lowering photosynthesis
D) High density reduces soil mineral concentration around roots

38. How does applying potassium fertiliser benefit tuber crops such as potato?

A) It enlarges the leaf blade area to increase total photosynthesis
B) It promotes assimilate transport to storage organs and increases starch
C) It reduces the light compensation point of the crop plant
D) It increases stomatal density on leaves to improve CO_2 uptake

39. Which temperature range is optimal for tuber formation in potato?

A) 5–10 °C
B) 10–15 °C
C) 18–23 °C
D) 25–30 °C

40. Ensuring adequate water supply to crops is important for photosynthesis because

A) water lowers leaf-cell temperature to the optimum level
B) water is a raw material of photosynthesis and shortage reduces efficiency
C) water increases CO_2 concentration dissolved inside the leaf
D) water activates PEP-carboxylase in C_4 and CAM species